

CI — The Chemical Industry

OCR B (Salters) H433 storyline **CI**. Nitric acid, sulfuric acid and ethanoic acid: kinetics and rate equations, equilibrium constants and their calculation, and choosing the best conditions for an industrial process.

1. Rate Equations and the Arrhenius Equation

Rate Equations

Rate equation: An expression showing how the rate of a reaction depends on the concentrations of reactants. For a reaction involving reactants A and B: $\text{rate} = k[A]^m[B]^n$, where k is the rate constant, m and n are the orders of reaction with respect to A and B, and the overall order is $m + n$.

The orders m and n can only be determined experimentally (they cannot be deduced from the balanced equation). Common orders:

Order	Effect of doubling [A]	Rate–concentration graph
0 (zero order)	Rate unchanged	Horizontal line (rate independent of [A])
1 (first order)	Rate doubles	Straight line through origin ($\text{rate} \propto [A]$)
2 (second order)	Rate quadruples	Upward curve ($\text{rate} \propto [A]^2$)

The rate constant k changes with temperature (increases as temperature increases) and is affected by catalysts, but is independent of concentration.

Determining Orders from Experimental Data

The **initial rates method** uses data from experiments where the initial concentration of one reactant is varied while others are held constant. Compare pairs of experiments to see how changing the concentration of one reactant affects the initial rate.

A **concentration-time graph** for a first-order reaction shows exponential decay. The half-life (the time for the concentration to halve) is constant for a first-order reaction. For zero order, the concentration decreases linearly with time. For second order, the concentration decreases as a reciprocal function.

Rate-Determining Step

Rate-determining step: The slowest step in a multi-step reaction mechanism. It acts as a bottleneck and determines the overall rate. The rate equation reflects the species involved in the rate-determining step and any preceding steps.

If the rate equation is $\text{rate} = k[A][B]$, then both A and B (or species derived from them) are involved in or before the rate-determining step. Species that appear in the balanced equation but not in the rate equation are involved in steps after the rate-determining step.

The Arrhenius Equation

$$k = Ae^{-E_a/RT}$$

where k is the rate constant, A is the pre-exponential factor (related to collision frequency and orientation), E_a is the activation energy (J mol^{-1}), $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, and T is temperature in K.

Taking natural logarithms gives the linear form:

$$\ln k = \ln A - E_a / RT$$

Plotting $\ln k$ against $1/T$ gives a straight line with gradient $= -E_a/R$ and y-intercept $= \ln A$. This allows E_a to be determined experimentally.

Exam Tip: The Arrhenius equation quantitatively explains why rate increases with temperature. A small increase in T produces a large increase in k because of the exponential relationship. When reading values from an Arrhenius plot, ensure you convert the gradient correctly: $E_a = -\text{gradient} \times R$, and remember units must be consistent (J not kJ for R).

Worked Example — Determining Order from Data:

Experiment 1: $[A] = 0.10$, rate $= 3.0 \times 10^{-4}$

Experiment 2: $[A] = 0.30$, rate $= 2.7 \times 10^{-3}$

$[A]$ is multiplied by 3 ($0.10 \rightarrow 0.30$). Rate changes by factor $2.7 \times 10^{-3} / 3.0 \times 10^{-4} = 9.0 = 3^2$

Therefore order with respect to A = **2** (second order).

2. Equilibrium Constants

The Equilibrium Constant K_c

For a general reaction: $aA + bB \rightleftharpoons cC + dD$

$$K_c = [C]^c[D]^d / [A]^a[B]^b$$

where square brackets denote equilibrium concentrations in mol dm^{-3} . K_c is constant at a given temperature. A large K_c means the equilibrium lies to the right (products favoured); a small K_c means it lies to the left (reactants favoured).

Only species in the aqueous or gaseous phase are included in the expression. Pure solids and pure liquids are excluded because their concentrations are effectively constant.

The Equilibrium Constant K_p

For gaseous equilibria, K_p is expressed in terms of partial pressures:

$$K_p = p_C^c \cdot p_D^d / (p_A^a \cdot p_B^b)$$

Partial pressure: The pressure that a gas would exert if it alone occupied the total volume of the mixture at the same temperature. The partial pressure of gas A is: $p_A = x_A \times P_{\text{total}}$, where x_A is the mole fraction of A.

Mole fraction: x_A = moles of A / total moles of all gases.

The total pressure equals the sum of all partial pressures (Dalton's law): $P_{\text{total}} = p_A + p_B + p_C + \dots$

Exam Tip: When calculating K_p , pay careful attention to units. The units of K_p depend on the stoichiometry of the reaction. Work out the overall power of pressure in the numerator minus the denominator to determine the unit (e.g. Pa, kPa, atm, or dimensionless).

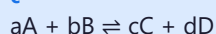
Equilibrium Constants — What K Tells Us

$K \ll 1$
Reactants favoured

$K \approx 1$
Neither side favoured

$K \gg 1$
Products favoured

K_c (concentrations)



$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

K_p (partial pressures)

For gaseous equilibria

$$K_p = \frac{p_C^c p_D^d}{p_A^a p_B^b}$$

What affects the value of K?

✓ Temperature changes K (shifts equilibrium in endothermic direction)

✗ Concentration, pressure, and catalysts do NOT change K